PERSONAL INFORMATION

# Kibrom Haftu

Arada, Addis Ababa, Ethiopia

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kibromhaftu (Skype)

Sex M | Date of birth 24/09/1988 | Nationality Ethiopian

## SUMMARY

As a passionate neural network (AI) programmer and an experienced Data & Web Engineer, I have a lot of experience in the field.

### Top Areas of Expertise: -

- Artificial Intelligence (Machine Learning, Deep Learning)
- Software Design & Development
- Data Mining, and Algorithms
- Managing Healthcare and financial systems (e.g., Core banking and imaging platforms)
- Risk and control frameworks (i.e., COBIT, ITIL, ISO27001) and ITGC best practices

## WORK EXPERIENCE

April 01, 2022 - Present

### Project Lead Data Scientist | Afro Chest X-ray Imaging Dataset

Main Responsibilities:

- Lead the development and implementation of the project. As the project lead, I ensure that the project is on track and meeting its goals by defining the project scope, timeline, and deliverables. This includes overseeing the collection and curation of the dataset, ensuring data quality, ensuring data privacy, developing data pre-processing pipelines, develop a standardized process (e.g., implement automated scripts using Python to extract metadata), building documentation, and
- Build Baseline AI System: This would involve designing the neural network architecture, selecting the appropriate activation functions, regularization techniques, and optimization algorithms. Using the data, the model will be trained, validated, fine-tuned, and tested.

July 01, 2022 - Present

### Manager -- Applications and Database Administration Division | LIB

Main Responsibilities:

- Lead in the development of standards to improve data integration between core operational and supported applications.
- Manage and mentor a team of applications and database administrators, providing technical guidance and leadership to promote a flexible and collaborative atmosphere.
- Ensure both teams follow all quality assurance processes.
- Deliver technology projects to meet business objectives and develop database architectures, coding standards, and quality assurance policies and procedures.
- Collaborate with others to implement Oracle High Availability Architecture (RAC, ASM, Flashback, Oracle Restart, Backup and Recovery plans at Datacentres as well as at Disaster Recovery Sites.

May. 2020 – June 2022

### Senior IT and System Auditor | Lion International Bank S.C. (LIB)

Main Responsibilities: -

- Carry out evaluations of the effectiveness of IT general controls (ITGC) and enterprise application controls (ITAC) such as Core Banking, Card Banking, and Mobile/Internet Solutions.
- Plan, organize, direct, manage, and conduct regular internal audits of information systems, applications, and the IT processes and auditing function of the bank.
- Make strong recommendations for control and IT process improvements.
- Report to the Board of Directors and Executive Management on audit status and results.

- Ensures alignment of internal processes with industry standards, expectations, and regulatory requirements.

Business (sector) Banking.

OCT. 2017 -DEC 2019

**Senior Database Administrator - DBA** | **K**ifiya **F**inancial **T**echnology PLC.

Main Responsibilities: -

- Conduct Strategic Project Implementations that enhances financial technologies
- Carry out requirement analysis and study of database and application side based on the organization requirement.
- Troubleshoot all database and application issues where the database administrator escalated.
- Implement database backup and recovery strategy.
- Supervise periodic database and application performance tuning and optimizations tasks
- Develop database and application scripts (Triggers, functions, packages, and procedures)

Business (sector) Financial Technology.

**Contributions**: - *Ensuring the continuation and enhancements of financial and payment services.*

OCT. 2014 –SEP. 2017

**Software Developer** | **B**oston **C**onsulting- **I**nfinity **A**dvanced **T**echnology PLC.

***Led team of developers***

- Responsible for designing, developing, and implementing Internet/Intranet applications.
- Determining design methodologies and toolsets; completing programming using languages and software products.
- Designed the architecture & led the implementation of the backend services
- Design, build and operate scalable implementations of data integrations

NOV. 2013-SEP 2014

**Information Technology Engineer** | Gibson School System.

Main Responsibilities: -

- Provided all levels of IT support for branch offices in networking, software, hardware, networks, and overall computer systems.
- Periodically provide reports to the managing director.
- Teaching Computer Programming to high school students.

JUN. 2012- JUL. 2013

**Project Lead** | Maichew Particle Board Manufacturing PLC.

Main Responsibilities and Roles: -

- Programmed and deployed a strong Web application that delivers numerous improvements to the human resource, marketing, supply, and procurement departments.
- Designed and Led the implementation of a software platform as per the requirements.
- Delivered High-quality Enterprise software.

Business or sector: Manufacturing

## EDUCATION AND TRAINING

SEP 2018- FEB 2021

### Addis Ababa University

Computer Science

**Masters of Science (MSc.) in Computer Science (Specialized in Data and Web Engineering)**

Grade (***CGPA***) of **3.85** on a 4.0 scale

**Courses: -**

- Research Methods (CoSc 6001)
- Computer Security (CoSc 6301)
- Data Mining and Data Warehouse (CoSc 6206)
- Distributed Systems (CoSc 6003)
- Software Project Management (CoSc 6101)
- Distributed Database Systems (CoSc 6201)
- Natural Language Processing (CoSc 6405)
- Multimedia and spatial-data management (CoSc 6202)
- Seminar in Data and Web Engineering (CoSc 6208)
- Digital Image Processing (CoSc 6012)
- Semantic Web Technologies (CoSc 6204)
- Pedagogy (CTPD 601)

**Research: -**

- ✓ Designed and programmed a novel Cardiorespiratory Diseases **Detection Algorithm** using Deep learning Algorithm.
- ✓ I have used Google developed **GCP** (Computing infrastructure) and Facebook developed **PyTorch** framework.

SEP. 2009- JUL. 2013
**(5 Years Study)**

## Mekelle Institute of Technology, MIT

Engineering and Technology

Awarded full tuition-free scholarship including full coverage of accommodation.

**Bachelor of Science (BSc.), Information Technology Engineering.**

Principal Subjects: -

- Data Structures (CSE 303)
- Design & Analysis of Algorithms (CSE 302)
- Advanced Database Programming (IT 302)
- Geographic Information System.
- Object-Oriented System Design (IT 405)
- Data Warehousing and Mining (IT 501)
- Distributed Database (IT 504)
- Web Design and Development (IT 204)
- Object-Oriented Programming with C++ (CSE 201)
- Rapid Application Development (IT 403)
- Mobile Computing (IT 408B)
- Software Engineering (CSE 304)
- Engineering Economics (GC 502)
- Data Warehousing and Mining (IT 501)
- Engineering Mathematics **I, II, III**
- Cryptography and Network Security (CSE 503)
- System Administration (IT 404)

SEP.2005-JUN. 2009

## Kellamino Special High School

Scholarship

Awarded full excellence scholarship including full Coverage of accommodation for four years.

Score: Very Great Distinction

NOV.2015-APR. 2016

## Oracle University

Boston Consulting

Studied and attended an approved oracle university hands-on class on: -

- Oracle Database: SQL, PL/SQL Fundamentals, and **Shell Programming**.
- Oracle database 11GR2: Data Guard Administration
- Oracle Database 11GR2: Real Application Clusters (RAC)
- Oracle Database 11g: Performance Tuning Release 2
- Oracle Database 11g: Administration I, II (Professional Level)
- Oracle WebLogic Server, Oracle Database 11g: Security
- Oracle 11g streaming

**Tools and Platforms used**: Oracle Enterprise Manager, SQL*Plus, Recovery

Manager (RMAN), data visualization tools (Toad for Oracle), SQL Developer, Oracle Linux Enterprise (a Redhat distribution Linux)

## PERSONAL SKILLS

**Languages: -**

- **English** – Writing, Reading & Speaking **Fluently**.
- **Amharic** – Writing, Reading & Speaking **Fluently**.
- **Tigrigna** – Writing, Reading & Speaking **Fluently**.

**Platforms: -**

- **UNIX or LINUX: - REDHAT Enterprise Linux, Ubuntu, Oracle Linux, Centos, Fedora, Linux Mint, Windows, Google Cloud Platform**

**Expertise: -**

- **Platforms – Django, Laravel (a PHP based Framework)**
- **Programming Languages – Strong Experience with Python: proficiency developing software and computer vision models, Java, C, C++, Matlab and solid experience applying statistical methods to data analysis using R programming.**
- **Libraries - Deep expertise in PyTorch, Tensorflow and similar packages**
- **Tools – Git, R Studio, RMAN, SQL Developer, TOAD.**
- **Storage – MYSQL, ORACLE DATABASE, SQL Server.**
- **Virtualization – Oracle VM SERVER, VIRTUALBOX, and VMware.**

**Independent Coursework**

- **Data Science Specialization [Coursera], Deep Learning Specialization [ Coursera] Machine Learning, Distributed Database Systems, Computer Vision, Data Engineering Nano Degree [Udacity], ISACA Audit Standards and Audit Methodology Principles, IT Governance using ITIL and COBIT, Machine Learning (ML) in the banking sector**

**Honours and Awards**

- **Fully funded Excellence Scholarship to join Kellamino Special High School.**
- **MIT, Mekelle Institute of Technology Excellence Scholarship Award.**

## ADDITIONAL INFORMATION

**Projects**

I worked on a lot of projects as a project manager and team member. The most notable ones are: -

- **Football Federation SMS Project**: - Implement Short Code SMS (8911).
- **Built Tikur-Chest X-ray Dataset for benchmarking AI models.**
- **Building Afro-chest X-ray Dataset** (**Ongoing project**)
- **Build a predictive data mining model for Oromia International Bank S.C.**: Firstly, a data warehouse solution is developed and then apply data mining algorithms to predict where the next branch should be opened based on key characteristics.

  **Lehulu:** - Participate in Lehulu platform (an Ethiopian Unified Billing system)
- **OCService**: - Online Customer Service for Ethiopian Companies with the capability to register, send complaints, requests, and orders!

- **iManage**: - Built Project management web application in partnership with 4 team members. Leveraged **LARAVEL** PHP framework and MYSQL for development.
- **Build Server Monitoring Tool**: - Based on Centos Operating System.
- **Ball Detection System**: Apply my machine learning skills in a real-world setting to accurately detect and track balls even in cluttered backgrounds.
- **Named Entity Recognition for Amharic using Gate toolkit:** This is an information extraction project for Ethiopic languages, mainly Amharic, the official language of Ethiopia, with the capability of extracting locations, persons, organizations, dates, times, and monetary amounts from documents using Jape programming.

## Research

**Cardiorespiratory Disease Detection using Deep Transfer Learning (DTL)**: - Designed and developed a novel nonparametric Bayesian factor model to neural network structure learning to achieve performance that outperforms the existing state-of-art (SOTA) approaches. The experiment was performed on Google Cloud Platform (GCP) using Python programming. The approach introduces a framework for data-driven transfer learning of useful features for cardiorespiratory disease detection by enabling learning the number of latent factors for nodes in a network's hidden layer that fit the available data.

## References

- Associate Prof. Dr. Yaregal Assabie (yaregal.assabie@yahoo.com or yaregal.assabie@aau.edu.et): who teaches me two courses: NLP, Computer Vision and **he was also my advisor**. Phone: +251(0)910914601
- Asst. Prof. Dr. Fekade Getahun (fekade.getahun@aau.edu.et or fekadegetahun@gmail.com): teaches me distributed database design, Data warehousing and data mining, Semantic Web Technologies. Phone: +251(0)911631615